A report on

Synthesis of Copper Doped Indium Sulfide for PEC Water Splitting



Prepared for

Dr. Sarbani Ghosh,

Assistant Professor, Department of Chemical Engineering

and

Dr. Mrinmoyee Basu,

Assistant Professor, Department of Chemistry

By:

Shivansh Sharma (2021A1TS2593P)

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Abstract

This work presents a comprehensive study on systematic transition from n-type to p-type semiconductor behavior of Cu-doped In_2S_3 crystals synthesized by a simple hydrothermal method using L-cysteine as the sulfur source. Field emission scanning electron micrograph revealed uniform rod-like shape with sub-micron size, providing enhanced surface area for catalytic reaction. X-ray diffraction analysis confirmed the cubic crystal structure of In_2S_3, with systematic peak shifts towards higher 2θ angles with increasing copper content, indicating successful substitutional doping. Inductively coupled plasma mass spectrometry revealed the actual copper incorporation, with approximately 8.6% doping in samples synthesized with 12% nominal concentration. Linear sweep voltammetry measurements revealed that bare In_2S_3 exhibits strong n-type behavior, as reflected in sharp photocurrent response (up to 3.0 mA/cm^2 at 0.6 V) in the anodic range and strong negative photocurrent (-0.7 mA/cm^2 at -0.8 V) in the cathodic range, with little dark current at positive potentials. With incorporation of Cu (4%, 6%, 8%, 10%, and 12%), progressive change in semiconductor characteristics was observed, with systematic suppression of photocurrent over the potential window, strong dark current emergence at elevated anodic potentials for heavily doped samples, and distinct shift in flat-band potential towards more negative potentials. Chronoamperometric measurement at -0.4 V confirmed this carrier-type conversion, with the photocurrent response decreasing considerably with higher Cu content. The 12% Cu-doped sample showed strong p-type behavior with little photocurrent response in the cathodic range and strong dark current in the anodic range. Density functional theory calculations provided theoretical explanations for the observed band gap narrowing and electronic structure changes resulting from copper doping. These results determine that controlled Cu doping in In_2S_3 crystals successfully modulates the electronic structure and majority carrier type, which changes from an intrinsically n-type semiconductor to a p-type semiconductor at elevated doping concentrations. This reversible carrier-type change provides significant potential for designing In_2S_3-based materials to suit advanced photoelectrochemical applications, optoelectronic devices, and photovoltaic systems.

Introduction

The increasing global demand for clean and renewable energy has driven the search for substitutes for fossil fuels, as concerns regarding environmental degradation and dwindling resources continue to rise. Of the numerous renewable energy sources, photoelectrochemical (PEC) water splitting is a potential technique for the production of hydrogen, a clean fuel, via solar energy-induced splitting of water to hydrogen and oxygen. The technique not only offers a solution to the global energy crisis, but also halts the adverse impacts of the use of fossil fuels, such as greenhouse gas emissions and climate change.

Critical to the advances in PEC water splitting is the development of highly efficient, stable, and cheap semiconductor materials to act as photoelectrodes. Indium sulfide (In_2S_3) has been a particularly promising candidate due to its suitable bandgap, favorable electronic properties, and chemical stability under operating conditions. Recent research has indicated that careful regulation of the crystal phase and morphology of In_2S_3 can significantly enhance its photoelectrochemical activity, enabling higher photocurrent densities and improved efficiency of charge separation. For instance, cubic-phase In_2S_3 nanostructures have exhibited top-notch photocatalytic behavior, and further modification such as cocatalyst deposition or heterostructure formation, has resulted in impressive improvements in solar-to-hydrogen conversion efficiencies.Despite these advances, numerous challenges remain to be overcome in order to optimize In_2S_3-based photoelectrodes for practical PEC water splitting. Charge carrier recombination, surface defect states, and the need for scalable synthesis pathways are some of the challenges that continue to limit overall device efficiency and stability. Solving these challenges requires a fundamental understanding of the interplay among crystal structure, morphology, surface chemistry, and photoelectrochemical response.

This thesis focuses on a systematical investigation of indium sulfide crystals as photoelectrodes for the splitting of water, with a special focus on how their structural characteristics engage their PEC activity. Through examinations of various synthesis pathways, crystal phases, and surface modifications, this research will aim to advance understanding of charge transport and catalytic mechanisms governing In_2S_3-based systems. Through insights that will be established, design of high-performance photoelectrodes will be rationalized to enable effective generation of clean hydrogen by PEC water splitting, a development which will open the opportunities for solar-powered PEC water splitting as a viable clean energy technology. Indium sulfide (In_2S_3) photocatalytic activity is a process of light energy being converted into chemical energy to drive a series of reactions, including pollutant degradation and photoelectrochemical water splitting to generate hydrogen.

In_2S_3 is a semiconductor with the appropriate bandgap (typically 1.57-2.12 eV, varying with synthesis conditions) to be able to absorb visible light energy. Some with compositions like β-In_2S_3 have bandgaps higher at ~2.67 eV. Upon exposure to visible light, In_2S_3 absorbs photons with energies higher than its bandgap, which excites electrons from the valence band (VB) to the conduction band (CB) and forms electron-hole pairs (e^-/h^+). The valence band (VB) position of In_2S_3 is typically 1.21 eV, and that of the conduction band (CB) is approximately -0.83 eV (ECB = E_g - EVB). In_2S_3 flat band potential (Vfb) depends on different electrolytes and has been found to be -1.2, -0.92, and -0.6 V vs. SCE with respect to different redox couples in solution.

The photogenerated holes (h^+) in the VB can oxidize water molecules to produce hydroxyl radicals (- OH):

$$h^+ + H_2O \rightarrow - OH + H^+$$

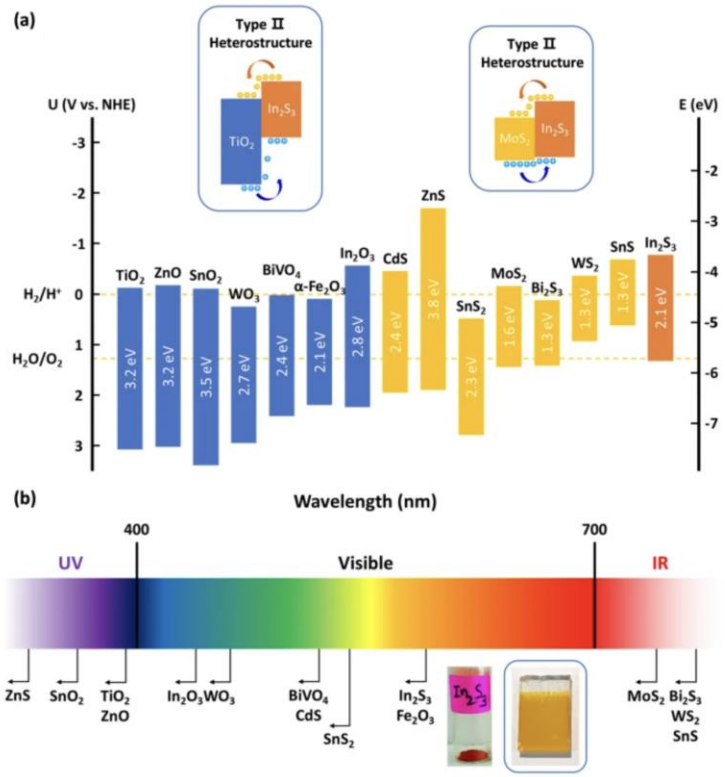


Figure 1. (a) Band edge positions of few metal oxides and metal sulfides using the normal hydrogen electrode (NHE) as a reference. Insets are examples of type II heterostructures with In2S3. **(b)** Electromagnetic wave spectrum and absorbable wave range of each semiconductor. Inset shows the color of In2S3. (Reproduced from https://doi.org/10.1039/c8ra09660k)

Additionally, the photogenerated electrons (e^-) in the CB can reduce oxygen or protons:

$$e^- + O_2 \rightarrow O_2^{-\,-}$$
$$2e^- + 2H^+ \rightarrow H_2$$

The position of the conduction band of In_2S_3 makes it thermodynamically favorable for hydrogen evolution, as its potential is more negative than the H^+/H_2 redox potential.
The photocatalytic performance of In_2S_3 can be significantly improved through two major ways. The first being heterostructure Formation. For instance, in Bi_2S_3-In_2S_3 heterostructures, Bi_2S_3 conduction band electrons are transferred into In_2S_3 valence band, which improves charge separation. Secondly, Doping with metals like Ti or V can introduce mid-gap intermediate bands, which can be used to introduce extra channels for photon absorption and increase the quantum efficiency by up to 50% for intermediate band solar cells. The redox and band structure of In_2S_3 make it an extremely promising material for a variety of photocatalytic applications, particularly if it is strategically designed to optimize its performance in photoelectrochemical devices.

Experimental Section

Synthesis of In_2S_3 and Cu-In_2S_3 Crystals

0.397 mmol $InCl_3 \cdot 3H_2O$ and 0.599 mmol L-Cysteine were weighed and dissolved in 15 ml solvent. Milli-Q water was employed in bare indium sulfide synthesis, whereas the lab-synthesized Cu stock solution was used for copper-doped samples. FTO glass substrate was cleaned thoroughly prior to both cleaned FTO and synthesized solution being placed in a hydrothermal flask. Hydrothermal reaction was conducted at 160 °C for 6 hours. Then, the FTO with synthesized nanoparticles was taken out of the reaction and dried in an oven at 40 °C for 1 hour. The dried sample was subsequently calcinated at 200 °C for 5 hours. Ultimately, the photoelectrochemical (PEC) activity of the prepared electrode was tested in 0.5 M Na_2SO_4 solution.

Characterisation

Crystallographic analysis was performed on an X-ray diffractometer with Cu Kα radiation (λ = 1.5406 Å). Scans were scanned from 10° to 80°. Morphological characterization was carried out in a field emission scanning electron microscope with high vacuum condition (1.10E-2 Pa). Micrographs were obtained in secondary electron (SE) mode in an Everhart-Thornley detector (ETD) of 32 μm aperture and 4.9 mm working distance. Micrographs were obtained at a magnification of 25000x, 50000x and 100,000x.

PEC measurements were conducted in three-electrode setup where In_2S_3 films synthesized were used as the working electrode, Ag/AgCl as the reference electrode, and platinum wire as the counter electrode, respectively. The electrolyte employed was 0.5 M aqueous Na_2SO_4 solution (neutral pH), and illumination was furnished by a Xenon arc lamp with a power of 100 mW/cm² for Photoelectrochemical Oxygen Evolution Reaction and Hydrogen Evolution Reaction measurements. All the chronoamperometric measurements were carried out at a constant potential of -0.4V with respect to Ag/AgCl reference electrode. EIS spectra were collected at constant DC potentials of 0 V and -0.4 V vs. Ag/AgCl, chosen from the LSV response regions of interest. Impedance data were acquired in the frequency range of 10^5 Hz to 0.1 Hz. Mott-Schottky measurements were carried out with the same three-electrode system and electrolyte as the EIS and PEC analysis. Capacitance-potential (C-V) correlations were measured using single-frequency impedance at 1 kHz with an AC perturbation amplitude of 10 mV. DC potential was swept between -1.0 V to +0.5 V vs. Ag/AgCl with a scan rate of 10 mV/s.

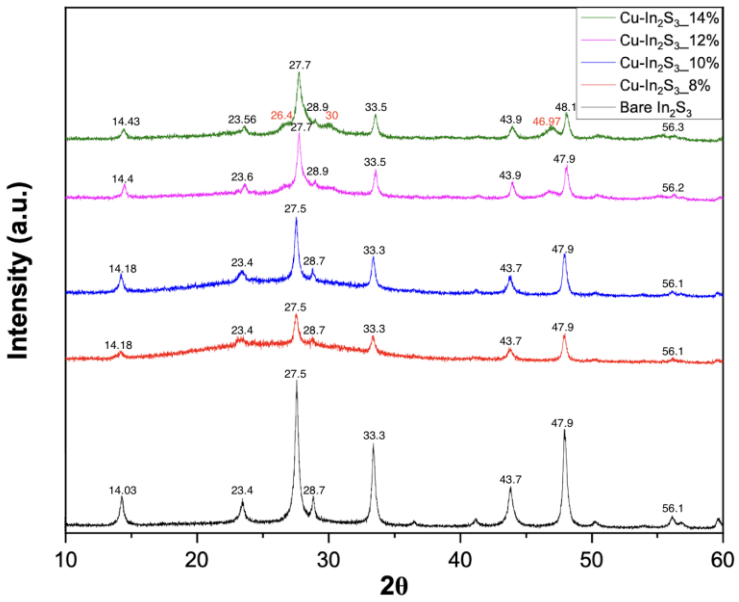


Figure 2. XRD patterns of BareIn2S3, and In2S3 doped with 8%,10%,12%,14% Cu Precursor with In precursor as basis.

Density Functional Theory Calculations

The crystal structure of In_2S_3 obtained from the XRD pattern was employed to construct a computational model for DFT calculations. A slab model was constructed atop the most prevalent crystallographic plane observed in the XRD pattern. All the DFT calculations were carried out with the Vienna Ab initio Simulation Package (VASP). The exchange-correlation functional was described with the Perdew-Burke-Ernzerhof (PBE) form of the generalized gradient approximation (GGA). A plane-wave cutoff energy of 520 eV was utilized, and the van der Waals interactions were treated by the DFT-D3 correction.

Geometry optimization of the bare In_2S_3 slab was performed with the conjugate gradient algorithm under ionic relaxation only but with fixed cell parameters. Electronic self-consistent field (SCF) convergence was at the level of 10^{-4} eV, up to 1000 electronic steps. Ionic convergence was at forces on all atoms less than 0.05 eV/Å. Partial occupancies were treated with a Gaussian smearing with a width of 0.05 eV. The Brillouin zone was sampled on a gamma-centered 2 × 2 × 1 k-point mesh.

Following geometry optimization of the undoped In_2S_3 slab, Cu atoms replaced In atoms in the structure at concentrations equivalent to the ICP-MS results. The Cu-doped system was then subjected to a second geometry optimization using the same computational settings. Electronic band structures and density of states were calculated for both optimized undoped and Cu-doped In_2S_3 systems to determine their respective band gaps and to examine the effect of Cu doping on the electronic properties of In_2S_3.

Results and Discussion:

Characterisation of In_2S_3 and Cu-In_2S_3 Crystals

XRD patterns indicate that copper doping is strongly correlated with peak positions across the series of undoped In_2S_3 to increasingly higher Cu-doped samples (8%, 10%, 12%, and 14%). With increasing concentrations of copper doping, there is a noticeable shift of diffraction peaks to higher 2θ angles, especially for the peaks at 27.5°, 33.3°, 43.7°, and 47.9°. Systematic increase to higher angles is indicative of a decrease in d-spacing by Bragg's law ($\lambda = 2d \cdot \sin\theta$), thus directly confirming lattice contraction. The Bare In2S3 observed its highest peak at 27.5°, corresponding to the [213] plane.

The contraction of the unit cell parameter observed is in agreement with the replacement of indium atoms by copper atoms, given that copper has a smaller atomic radius (128 pm) compared to indium (156 pm). Replacing a smaller atom for a larger one in a crystal lattice leaves less space requirement, leading to the observed lattice contraction. This gradual change is intensified with the content of copper, affirming the successful substitution of copper into the In_2S_3 crystal structure instead of the development of a second phase.

Notably, when copper doping is 14%, three extra diffraction peaks are seen at 26.4°, 30° and 46.97° (marked as red in the pattern), which are related to the presence of a distinct Cu indium sulfide phase. This shows that 12% is a critical concentration above which copper no longer merely occupies the native In_2S_3 lattice but begins to form as a distinct crystalline phase within the material.

Figure {} presents FESEM micrographs of the hydrothermally synthesized In_2S_3 sample at different magnifications. Micrographs reveal a three-dimensional hierarchical structure composed of interconnected nanoflakes as the predominant component. At low magnification (25,000×), the nanoflakes appear to be having a network structure with large void spaces between them, indicating a sponge-like structure. Micrographs at higher magnifications provide a clearer view of the individual nanoflakes, which possess curved morphology with thicknesses estimated to be in the range of 15-30 nm. The nanoflakes are randomly oriented and interconnected and create numerous channels and pores within the structure. The resultant nanoflake morphology is particularly useful for photoelectrochemical reactions. The hierarchical structure possesses several beneficial features: (1) significantly enhanced specific surface area, with high density of photocatalytic reaction sites; (2) improved light-harvesting capability through repeated reflections of light in the porous network; (3) decreased path lengths for charge carrier diffusion to the electrode-electrolyte interface; and (4) easier penetration of the electrolyte through the entire film. The open structure of the nanoflakes can also be used to allow fast electron transport within the material, with potential to reduce recombination losses.

This distinctive morphology is probably the outcome of the synergistic action of the hydrothermal synthesis condition and interaction between In^{3+} ions and L-cysteine, both as a sulfur source and as a structure-directing agent. Such nanoflake morphologies have previously been found to be linked with enhanced photoelectrochemical performance in In_2S_3-based systems because of the optimal balance between high surface area and efficient charge transport channels.

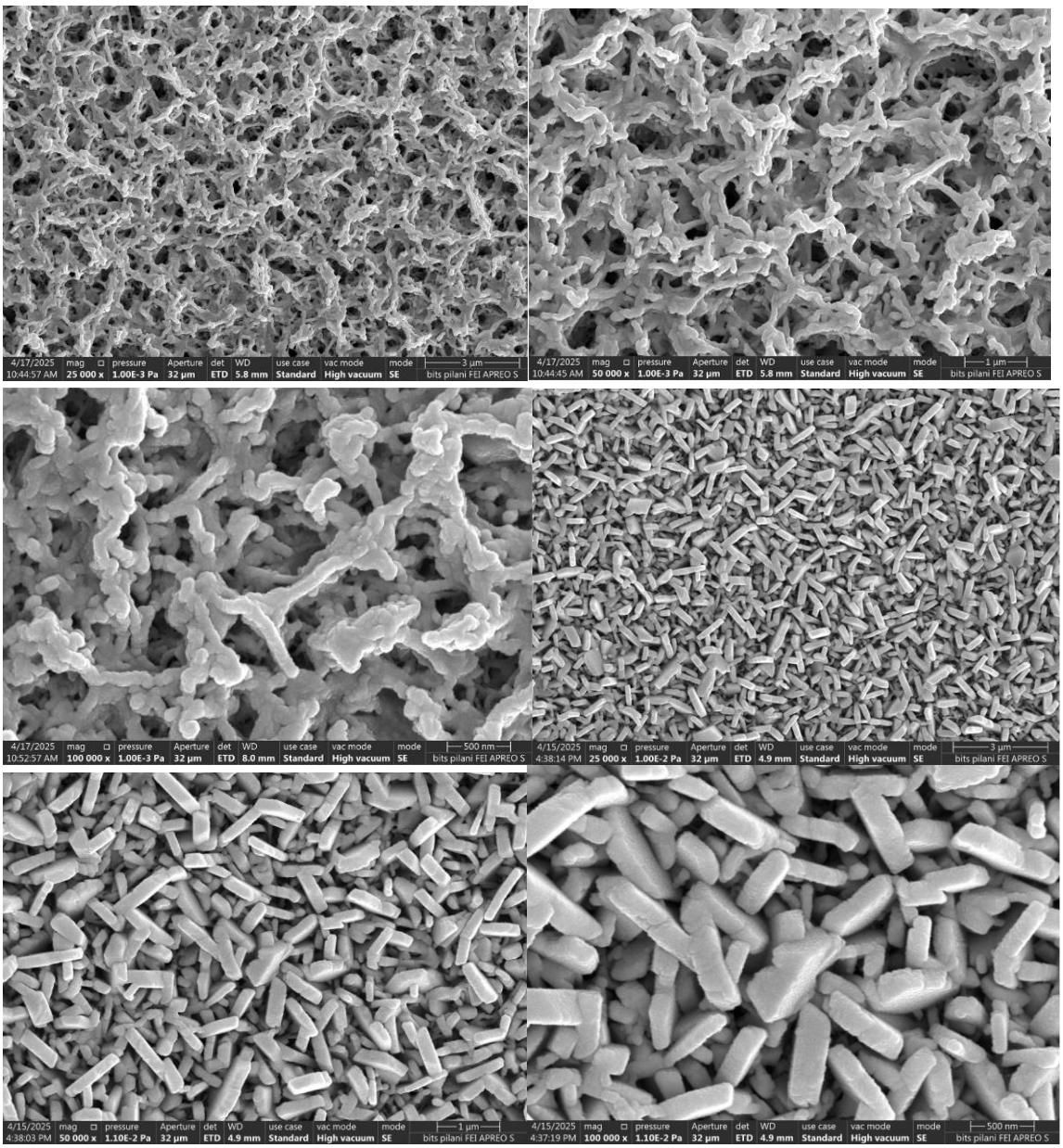


Figure 3. FESEM micrographs of an (a),(b),(c) Bare In_2S_3 and (d),(e),(f) 12% Cu Doped In_2S_3 thin film prepared on FTO substrate

FESEM characterization of the 12% Cu-doped In2S3 system indicates a clear morphological architecture made up of evenly distributed sheet-shaped nanostructures of about 100-200 nm in length and 30-50 nm in width. The nanostructures are well-defined in shape and are randomly oriented, and they form an interconnected three-dimensional network with big void spaces among neighboring nanosheets. Such a specific morphology provides a high surface-to-volume ratio configuration that indicates high surface reactivity potential. Homogeneity of nanosheet dimensions and distribution at different magnifications (25,000×, 50,000×, and 100,000×) verifies homogeneous copper dopant incorporation in the In2S3 matrix. The anisotropic growth pattern noticed is most likely due to the preferential growth of the crystals along some of the crystallographic directions due to the 12% copper doping, which has apparently radically altered the characteristic morphology of undoped In2S3.

Inductively Coupled Plasma Mass Spectrometry (ICP-MS) was used to measure the actual copper doping concentration in the as-synthesized Cu-In_2S_3 samples. Standard calibration

solutions of Cu with concentrations of 0, 1, 3, 5, 7, 9, and 11 mg/L were prepared and analyzed to obtain a linear calibration curve. Measurements were conducted at six different emission wavelengths typical of copper: 327.393, 324.752, 224.700, 213.597, 222.778, and 221.459 nm to provide analytical precision. The calibration standards showed excellent linearity ($R^2 > 0.9$) over the concentration range for all the measured wavelengths.

The test sample doped at a nominal level of 12% copper was measured under the same conditions. The measured concentrations of copper at detection were 9.15, 9.139, 8.89, 8.734, 7.278, and 8.363 mg/L for corresponding wavelengths. The results verify the actual copper incorporation in the Cu-In_2S_3 sample is below the precursor concentration, which was 8.59 ± 0.72 mg/L. This is approximately a copper incorporation efficiency of about 71.6%. Variation among different emission lines may result from potential matrix effects or minute spectral interferences. The most sensitive and repeatable wavelengths among Cu 327.393 and Cu 324.752 indicated that these are the best copper quantification emission lines for such samples. This analytical validation of copper content corroborates the structural change shown in the XRD patterns, wherein peak shifts revealed successful incorporation of copper into the In_2S_3 crystal structure.

Similarly, ICP-MS was carried out on the 12% Cu-In2S3 sample. A calibration curve was obtained using standard solutions of indium with different concentrations of 0, 5, 10, 20, 30, 35, and 40 mg/L. Four emission wavelengths shared by indium at 230.606, 325.609, 303.936, and 451.131 nm wavelengths were employed for analysis to make analysis precise and reproducible.The calibration standards showed perfect linearity ($R^2 > 0.9$) over the range of concentrations at all the measurement wavelengths. The test sample was measured under the same instrumental conditions, and the resulting indium concentrations were 54.964, 52.82, 55.023, and 56.675 mg/L for the respective wavelengths. The resulting values confirm that the indium content of the sample was greater than the upper calibration standard (40 mg/L), indicating a very high incorporation efficiency of indium precursors in the hydrothermal synthesis process. The mean indium concentration was calculated to be 54.87 ± 1.59 mg/L.

The comparatively low standard deviation (about 2.9% of the mean value) for the four emission lines testifies to outstanding analytical accuracy and suggests little spectral interferences in the measurement. The uniformly high indium composition is in agreement with the morphological and structural properties previously found in FESEM and XRD measurements, confirming the successful synthesis of the target sheet-like indium-rich In_2S_3 nanostructures. The high indium composition probably is the origin of the desirable electronic properties found in the photoelectrochemical performance tests.{UV-Vis}

Table 1: ICP-MS result of Cu in the 12% Copper Precursor doped Indium Sulfide crystals

Cu% in Sample Fed	Cu 327.393 (mg/L)	Cu 324.752 (mg/L)	Cu 224.700 (mg/L)	Cu 213.597 (mg/L)	Cu 222.778 (mg/L)	Cu 221.459 (mg/L)
0%	0	0	0	0	0	0
1%	1	1	1	1	1	1
3%	3	3	3	3	3	3
5%	5	5	5	5	5	5
7%	7	7	7	7	7	7
9%	9	9	9	9	9	9
11%	11	11	11	11	11	11

Cu% in Test Sample	9.15	9.139	8.89	8.734	7.278	8.363

Photocurrent Analysis

Linear sweep voltammetry (LSV) sweeps were carried out to study the photoelectrochemical activity of Cu-doped and bare In_2S_3 crystals. Fig. {} is photocurrent density vs. applied potential plot for bare In_2S_3 and Cu-doped samples of varied Cu concentration (4%, 6%, 8%, and 12%). Examination of the anodic sweep (majorly positive potentials) of LSV scans shows clear evidence of n-type to p-type semiconductor character transition with the increase in Cu doping concentration in Cu-doped In_2S_3 crystals. As Figures 1 and 2 show, undoped In_2S_3 exhibits strong photocurrent response under illumination in the anodic region, up to about 3.0 mA/cm^2 at 0.6 V, with very little dark current even at high positive potentials. This is typical of the intrinsic n-type character of In_2S_3, where photogenerated electrons dominate the measured photocurrent and dark carriers are minimal. In comparison, Cu-doped samples exhibit a declining photocurrent response with increasing Cu concentration and a corresponding increase in dark current. Of specific interest is that the higher Cu content samples (10% and 12%) possess enormous dark current at anodic potentials greater than 0.4 V. The dark current contribution is increasingly dominant with rising Cu doping level and possesses the maximum anodic-region dark current in the 12% Cu-doped sample.

Appearance of intense dark current at high positive voltages is unambiguous evidence for p-type semiconductor behavior in highly Cu-doped samples. This feature arises since Cu dopants introduce acceptor states in the bandgap of In_2S_3 and thus effectively introduce holes as majority carriers. These majority carriers are responsible for the dark current under anodic bias conditions in the absence of photoexcitation, which determines the onset of p-type conductivity.In addition, the systematic enhancement of dark current with Cu content is consistent with a prediction trend towards the conversion of the anticipated n-type to p-type behavior. The 12% Cu-doped sample shows little n-type character, as indicated by the very weak photocurrent response of the sample in the cathodic region and high dark current in the anodic region.

As for the cathodic sweep, bare In_2S_3 exhibits characteristic n-type semiconductor response with high negative photocurrent (-1.25 mA/cm^2) at cathodic potentials (-0.8 V) and high positive photocurrent (in excess of 1.1 mA/cm^2) at anodic potentials (0.2 V). High photocurrent response in both cathodic and anodic areas is an indicator of efficient charge collection and separation with electrons as majority carriers.

Table 2: ICP-MS result of In in the 12% Copper Precursor doped Indium Sulfide crystals

In% in Sample Fed	In 230.606 (mg/L)	In 325.609 (mg/L)	In 303.936 (mg/L)	In 451.131 (mg/L)
0%	0	0	0	0
5%	5	5	5	5
10%	10	10	10	10
20%	20	20	20	20
30%	30	30	30	30
35%	25	25	25	25

40%	40	40	40	40
In% in Test Sample	54.964	52.82	55.023	56.675

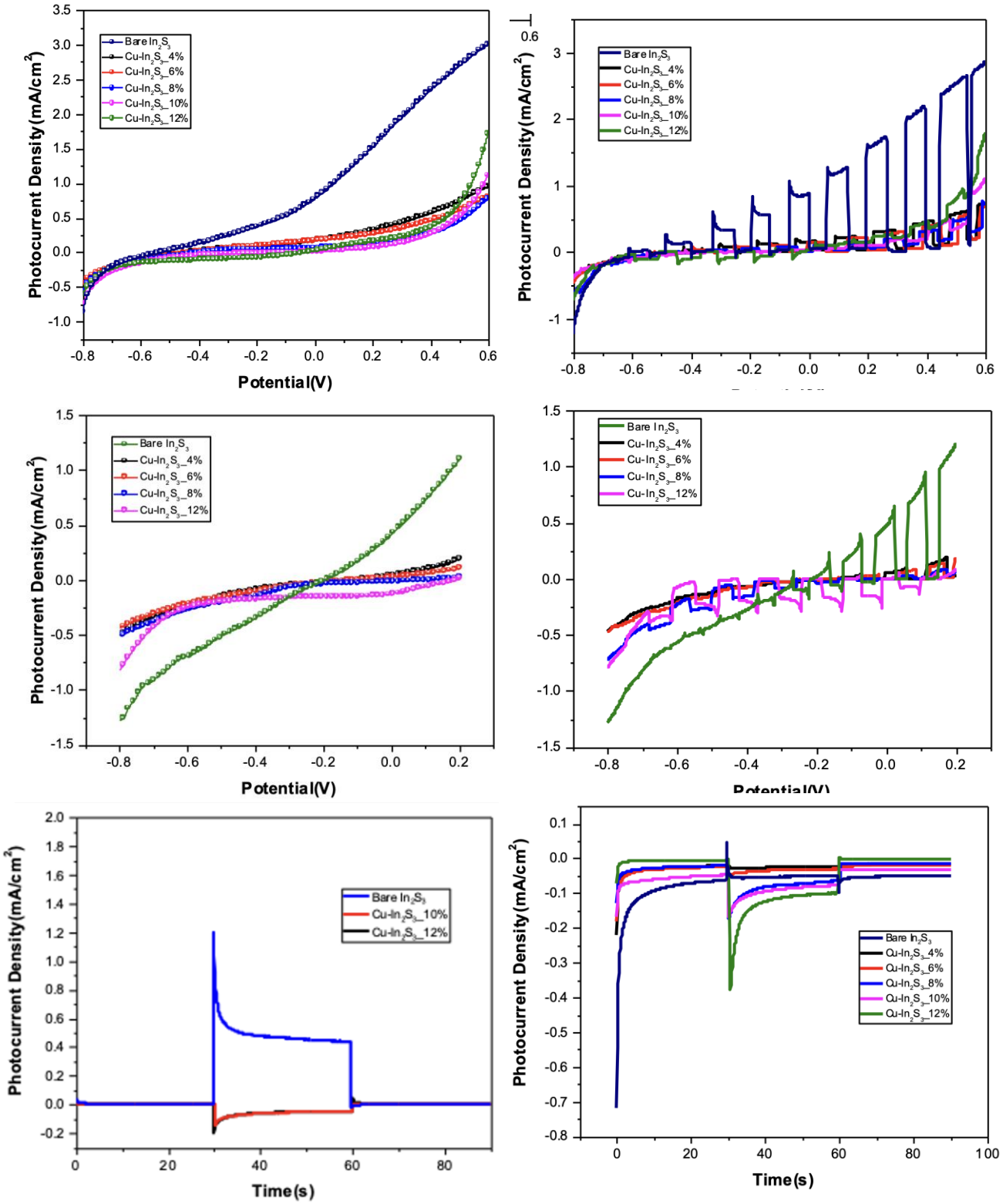


Figure 4. Photocurrent density–voltage (I–V) curves under simulated sunlight illumination(400nm) following **(a,b)** Anodic Sweep and **(c,d)** Cathodic Sweep,Photocurrent density–time (I–T) curvesmeasured at **(e)** 0V vs RHE, **(f)** -0.4V vs RHE under chopped

simulated sunlight illumination.

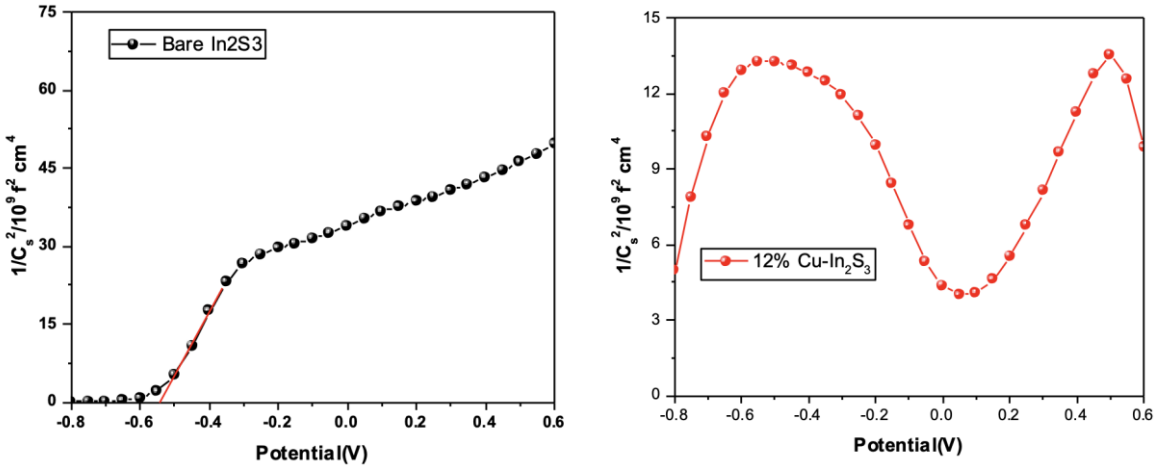


Figure 5. Mott-Schottky collected at the fixed frequency of {}

Bare In_2S_3 is especially found to respond with high dark current at negative potentials and zero dark current even at high positive potentials, further supporting its intrinsic n-type semiconductor nature. This is due to sulfur vacancies being electron donors in the In_2S_3 matrix.

With Cu addition, the photoelectrochemical response becomes systematic. The photocurrent-potential curves become more flat with higher Cu content, exhibiting a fundamental change in semiconductor behavior. At 4% Cu doping, the anodic photocurrent is reduced to around 0.2 mA/cm² at 0.2 V, and the cathodic photocurrent is reduced to -0.5 mA/cm² at -0.8 V. This trend continues with higher concentrations of Cu, with 12% Cu-doped In_2S_3 exhibiting very little photocurrent response across the entire potential window (<0.05 mA/cm² at 0.2 V and -0.7 mA/cm² at -0.8 V). A notable variation of flat-band potential is seen with rising concentration of Cu. The potential at which photocurrent goes through zero changes systematically towards the negative with rising Cu content, and it provides strong evidence of the transition from n-type to p-type semiconductor behavior. The cause of such transition is that Cu dopants create acceptor levels in the bandgap of In_2S_3, which

increasingly compensate the native n-type behavior and ultimately cause p-type conductivity at high doping levels. The 12% Cu-doped sample shows strong p-type behavior, with strongly suppressed photocurrent response over the whole range of potentials and a considerably different photocurrent onset than bare In_2S_3. This shows that holes are now the predominant charge carriers, taking over from electrons in the transport process. The modified band structure and charge transport dynamics caused by high Cu incorporation result in strongly dissimilar photoelectrochemical behavior than the bare n-type material.

Chronoamperometric responses at -0.4 V and 0 V under chopped illumination further elucidate carrier-type transition in Cu-doped In_2S_3 crystals. At -0.4 V, bare In_2S_3 exhibits a sharp negative photocurrent spike on illumination, typical of efficient electron transport and typical n-type semiconductor behavior, while enhanced Cu doping causes systematic diminishment and eventual absence of the anodic response, with the 12% Cu-doped sample exhibiting p-type

characteristics. At 0 V, bare In_2S_3 exhibits a large positive photocurrent, again typical of n-type character; however, the 10% Cu-doped sample exhibits an initial cathodic (negative) spike followed by a small positive steady-state current, indicating the coexistence and competition of n-type and p-type transport processes at intermediate doping

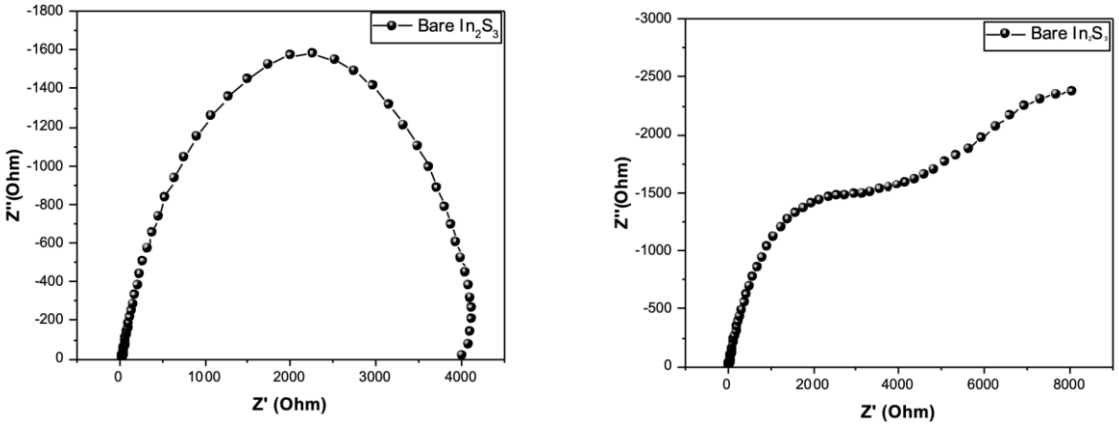


Figure 6. Electrochemical impedance spectroscopy carried out on Bare In_2S_3

concentrations. Of especial interest, the 12% Cu-doped sample exhibits essentially no photocurrent at 0 V, as would be expected in the case of extensive carrier recombination and fully developed p-type character. These results, together with LSV data, confirm that increased Cu incorporation brings about a systematic conversion from n-type to p-type conductivity in In_2S_3, with the transition being most clearly seen in the complete elimination or inversion of photocurrent responses at higher doping concentrations.

Mott-Schottky (M-S) analysis was used to further support the semiconductor type transition and assess the flat-band potentials of bare and Cu-doped In_2S_3 electrodes. Fig. 4 shows the M-S plots ($1/C^2$s versus applied potential) for bare In_2S_3 and 12% Cu-doped In_2S_3, recorded in 0.5 M Na_2SO_4 electrolyte. Bare In_2S_3 shows a clearly defined positive slope in the potential range from -0.4 V to 0.5 V vs. Ag/AgCl, unequivocally establishing n-type semiconductor behavior. The positive slope is characteristic of electron majority carriers, as predicted by the intrinsic nature of In_2S_3 due to sulfur vacancies. Extrapolation of the linear region to the x-axis gives a flat-band potential of approximately -0.6 V, which is the potential at which band bending is suppressed at the semiconductor-electrolyte interface. This contrasts starkly to the 12% Cu-doped In_2S_3 having a completely different M-S response of a deeply U-shaped plot with positive as well as negative slopes. Foremost, is a clear evidence of a distinct negative slope observed between the range of potential of -0.15 V and 0.1 V and hence definitive confirmation of p-type semiconductor behavior. The negative slope is indicative of the accumulation of holes in space charge region thus validating holes to be majority carriers. The flat-band potential calculated from the region of negative slope is around 0 V, being a large anodic shift with respect to native In_2S_3.

The multifaceted M-S behavior of the 12% Cu-doped sample, with additional positive slopes at both more negative and more positive potentials, indicates the existence of several electronic states or a heterogeneous surface composition. This complexity is probably due to Cu ions residing in different crystallographic positions in the In_2S_3 lattice, forming multiple acceptor levels in the band structure. However, the prevailing negative slope around 0 V definitively

proves the successful conversion from n-type to p-type semiconductor activity by deliberate Cu introduction. These M-S results corroborate and quantitatively substantiate the carrier-type transition in LSV and chronoamperometry measurements, confirming a thorough characterization of the electronic structure change induced by Cu doping in In_2S_3 crystals.

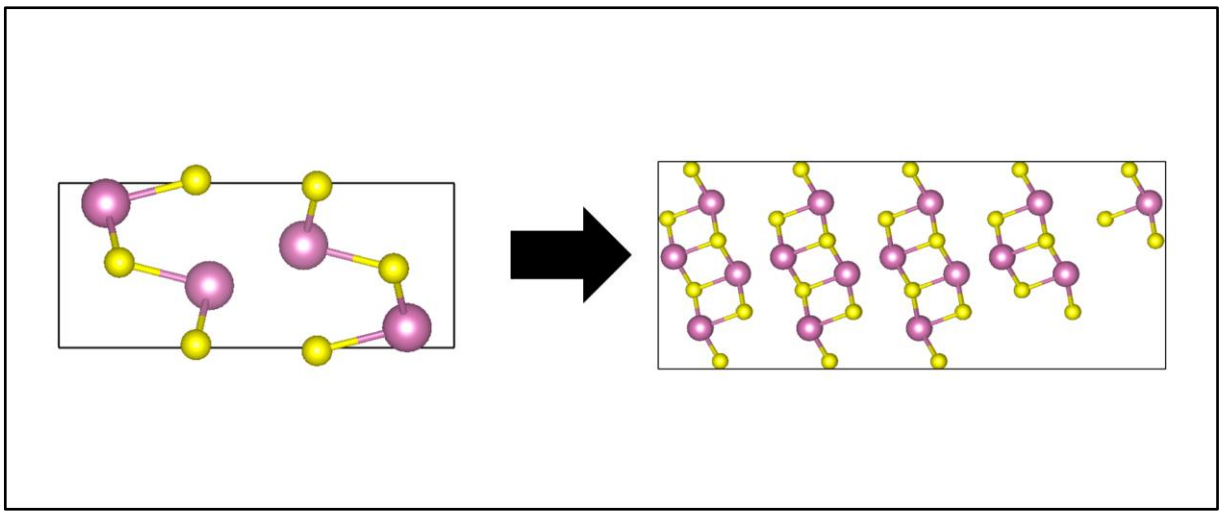


Figure 7. Conversion of In2S3 primitive unit cell to (213) Slab.

Electrochemical impedance spectroscopy (EIS) measurements of the bare In_2S_3, as represented in the Nyquist plots at 0V (left) and -0.4V (right) vs. Ag/AgCl, are highly informative of the interfacial charge transfer behavior under various bias conditions. Under 0V, the Nyquist plot has a nearly perfect semicircular shape with a diameter of around 4000 Ω, representing a moderate charge transfer resistance (Rct) at the electrode–electrolyte interface. This semicircular response implies that, at the flat-band potential, the charge transfer reaction is quite easy and largely controlled by surface kinetics as opposed to bulk diffusion, in line with efficient electron transfer for n-type In_2S_3.

As opposed to the above, the EIS spectrum at -0.4V displays a far larger, incomplete semicircle with a distinctive low-frequency tail, extending into approximately 8500 Ω along the real axis. This rise in both the semicircle diameter and the appearance of a Warburg-type diffusion tail is an indicator of a significant increase in charge transfer resistance and the beginning of diffusion-limited processes under cathodic bias. The increased impedance at -0.4V is due to enhanced band bending in the n-type semiconductor, which creates an extended depletion region and impairs electron transport through the interface.

Density Functional Theory Calculations on prepared In2S3

X-ray diffraction (XRD) analysis of the as-synthesized samples also established the existence of the β-In_2S_3 phase, with the diffraction pattern exhibiting prominent peaks of this structure. Interestingly, the (213) reflection was one of the strongest, suggesting a preferential orientation or extensive exposure of the (213) planes in the crystallites. This experimental finding gave a compelling reason for the development of a theoretical slab model from the (213) surface for

future computational investigation.

To accurately model the experimentally determined surface, a β-In_2S_3 unit cell was first constructed based on crystallographic parameters consistent with the XRD data. The unit cell, with its vacancy-ordered tetragonal structure, was then cleaved along the (213) plane to yield a slab model of the most exposed surface in the experimental samples.This is illustrated in Figure X, which illustrates the progression from the bulk unit cell to the (213) slab. The resulting slab maintains the intrinsic characteristics of the β-In_2S_3 structure, including the ordered indium vacancies and the stacking sequence of the atomic layers.

Conclusions

In this research work, we were able to clearly show that the intrinsically n-type In_2S_3 crystals have been converted to p-type semiconductors with controlled doping using copper. The rod-type nature of materials produced, verified using FESEM characterization, furnished additional surface area of value to the photoelectrochemical operations. XRD results confirmed β-In_2S_3 phase, but with orientation preferentially parallel along the plane (213), directing our calculation research through a correct slab model selection. Systematic conversion to p-type character was extensively elucidated through several complementary methodologies. Linear sweep voltammetry scans demonstrated increasingly complete suppression of photocurrent with elevated Cu composition, concomitant with considerable dark current development at anodic biases in heavily doped materials-a characteristic signature of p-type conduction. Chronoamperometry measurements at -0.4V and 0V also confirmed this transition, reporting the photoresponse inversion and eventual n-type carrier transport pathway suppression at 12% Cu doping. The structural-optical-photoelectrochemical property correlation developed in this work offers important insights into carrier-type modulation governing mechanisms in In_2S_3 via targeted dopant addition.

These results illustrate that Cu-doped In_2S_3 is a highly promising material system with tunable electronic properties for next-generation applications in photocatalysis, optoelectronics, and energy conversion technologies

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